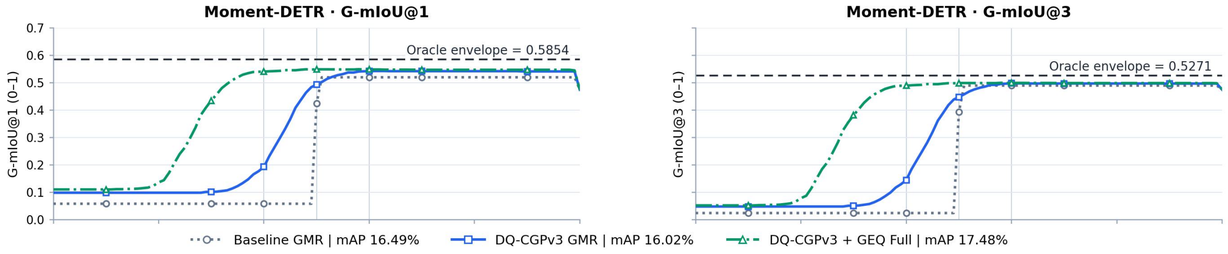
SoccerGMR G-mIoU Sensitivity by Backbone · Standard-trained → Standard Test (1036 queries)

Comparing Baseline GMR vs DQ-CGPv3 GMR vs DQ-CGPv3 + GEQ Full

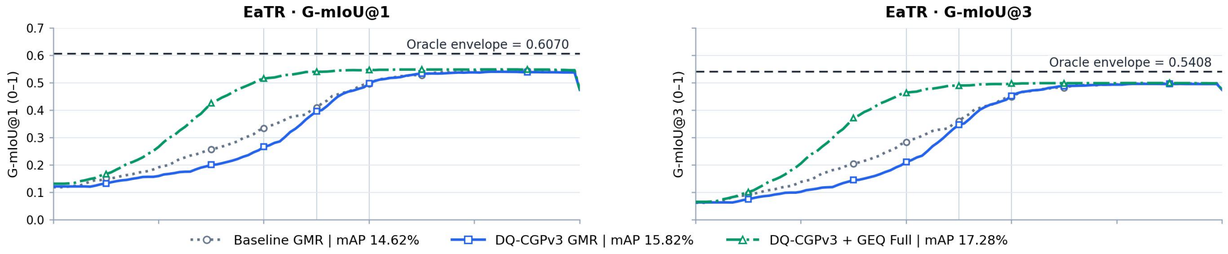
Baseline GMR | mAP 6.68% · DQ-CGPv3 GMR | mAP 10.66% · DQ-CGPv3 + GEQ Full | mAP 12.64%

ΔmAP (DQ-CGP vs Base): +3.98 pp · ΔG1 mean (.4/.5/.6): +7.54 pp · ΔG3 mean (.4/.5/.6): +6.10 pp



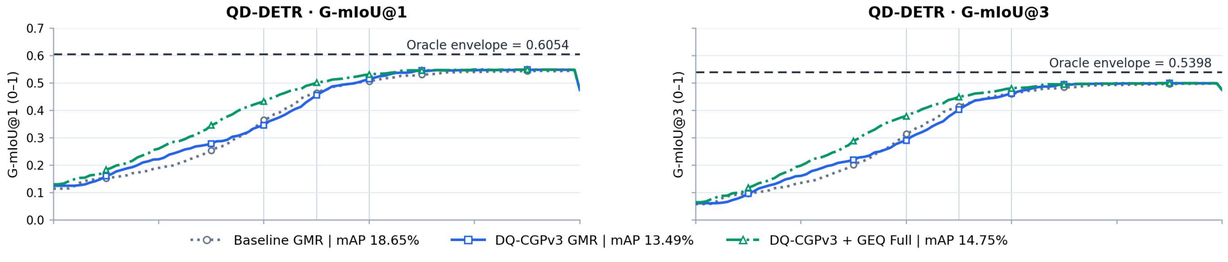
Baseline GMR | mAP 16.49% · DQ-CGPv3 GMR | mAP 16.02% · DQ-CGPv3 + GEQ Full | mAP 17.48%

ΔmAP (DQ-CGP vs Base): -0.47 pp · ΔG1 mean (.4/.5/.6): -2.63 pp · ΔG3 mean (.4/.5/.6): -2.70 pp



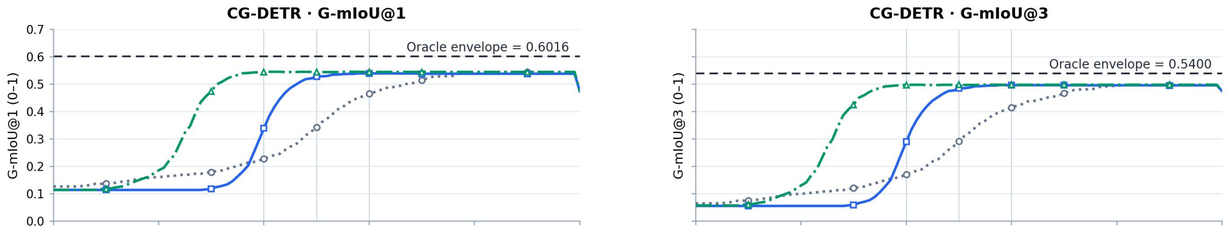
Baseline GMR | mAP 14.62% · DQ-CGPv3 GMR | mAP 15.82% · DQ-CGPv3 + GEQ Full | mAP 17.28%

ΔmAP (DQ-CGP vs Base): +1.20 pp · ΔG1 mean (.4/.5/.6): -0.62 pp · ΔG3 mean (.4/.5/.6): -1.08 pp



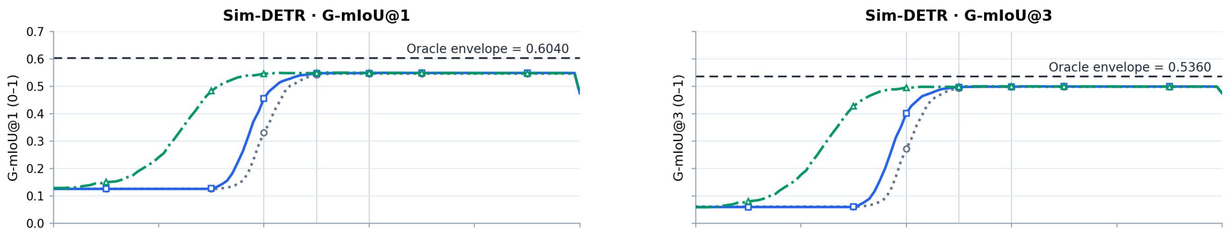
Baseline GMR | mAP 18.65% · DQ-CGPv3 GMR | mAP 13.49% · DQ-CGPv3 + GEQ Full | mAP 14.75%

ΔmAP (DQ-CGP vs Base): -5.16 pp · ΔG1 mean (.4/.5/.6): +12.31 pp · ΔG3 mean (.4/.5/.6): +13.14 pp



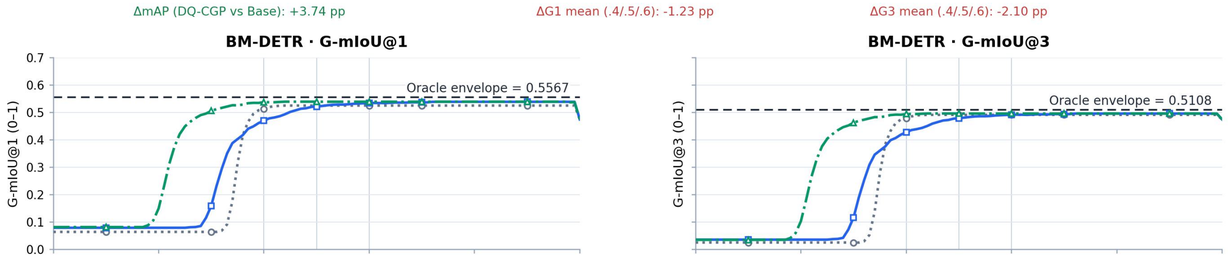
Baseline GMR | mAP 20.73% · DQ-CGPv3 GMR | mAP 20.63% · DQ-CGPv3 + GEQ Full | mAP 21.43%

ΔmAP (DQ-CGP vs Base): -0.10 pp · ΔG1 mean (.4/.5/.6): +4.39 pp · ΔG3 mean (.4/.5/.6): +4.45 pp



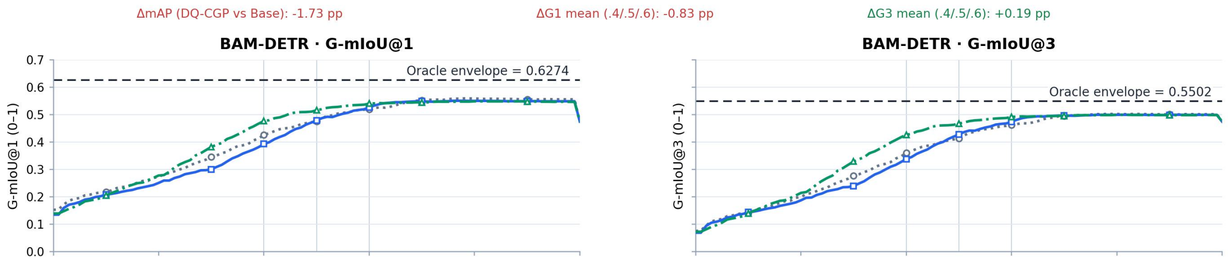
Baseline GMR | mAP 4.00% · DQ-CGPv3 GMR | mAP 7.74% · DQ-CGPv3 + GEQ Full | mAP 7.82%

ΔmAP (DQ-CGP vs Base): +3.74 pp · ΔG1 mean (.4/.5/.6): -1.23 pp · ΔG3 mean (.4/.5/.6): -2.10 pp



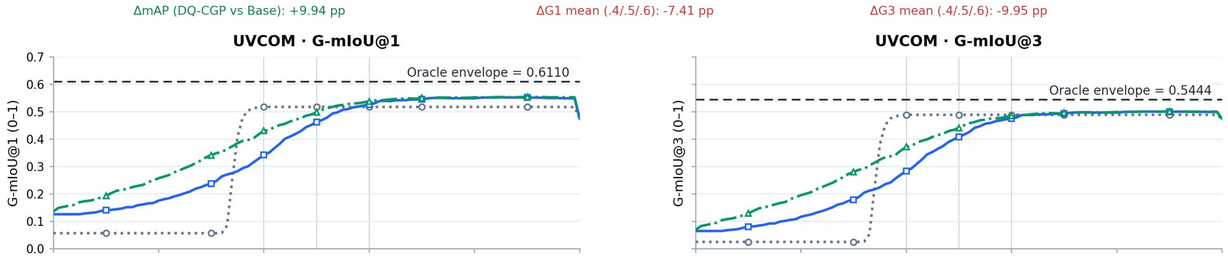
Baseline GMR | mAP 23.25% · DQ-CGPv3 GMR | mAP 21.52% · DQ-CGPv3 + GEQ Full | mAP 22.84%

ΔmAP (DQ-CGP vs Base): -1.73 pp · ΔG1 mean (.4/.5/.6): -0.83 pp · ΔG3 mean (.4/.5/.6): +0.19 pp



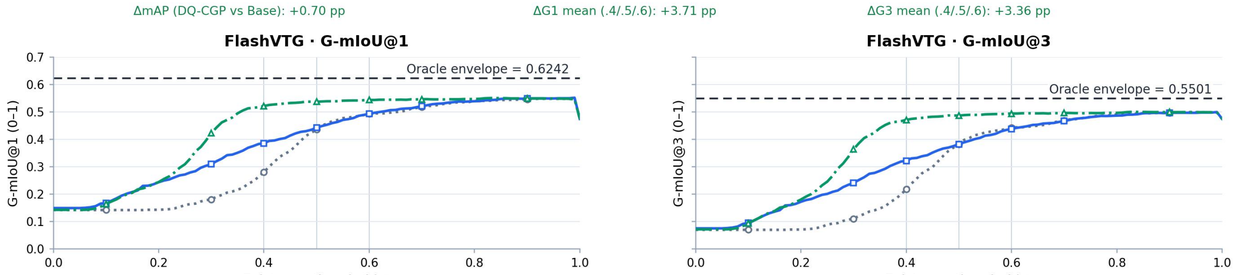
Baseline GMR | mAP 7.12% · DQ-CGPv3 GMR | mAP 17.06% · DQ-CGPv3 + GEQ Full | mAP 18.40%

ΔmAP (DQ-CGP vs Base): +9.94 pp · ΔG1 mean (.4/.5/.6): -7.41 pp · ΔG3 mean (.4/.5/.6): -9.95 pp



Baseline GMR | mAP 25.43% · DQ-CGPv3 GMR | mAP 26.13% · DQ-CGPv3 + GEQ Full | mAP 25.81%

ΔmAP (DQ-CGP vs Base): +0.70 pp · ΔG1 mean (.4/.5/.6): +3.71 pp · ΔG3 mean (.4/.5/.6): +3.36 pp



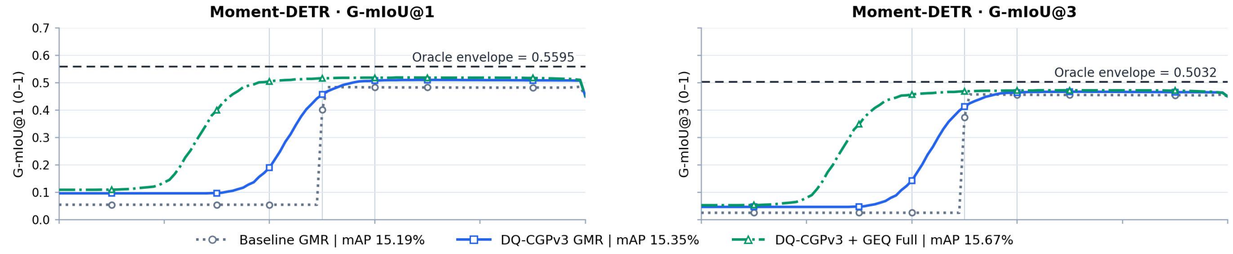
Oracle envelope: max of plotted methods with GT presence and fixed candidates. Guides at tau in {0.4, 0.5, 0.6}.

SoccerGMR G-mIoU Sensitivity by Backbone · Standard-trained → Full-Decodable Test (2970 queries)

Comparing Baseline GMR vs DQ-CGPv3 GMR vs DQ-CGPv3 + GEQ Full

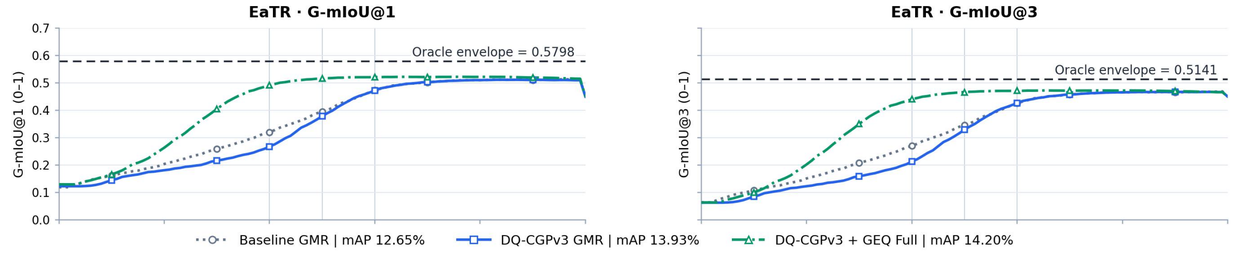
Baseline GMR | mAP 5.47% · DQ-CGPv3 GMR | mAP 9.99% · DQ-CGPv3 + GEQ Full | mAP 11.63%

ΔmAP (DQ-CGP vs Base): +4.52 pp · ΔG1 mean (.4/.5/.6): +7.19 pp · ΔG3 mean (.4/.5/.6): +5.48 pp



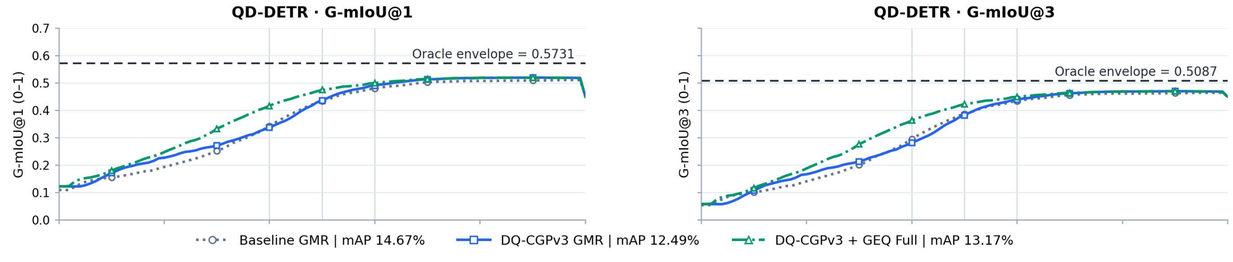
Baseline GMR | mAP 15.19% · DQ-CGPv3 GMR | mAP 15.35% · DQ-CGPv3 + GEQ Full | mAP 15.67%

ΔmAP (DQ-CGP vs Base): +0.16 pp · ΔG1 mean (.4/.5/.6): -2.34 pp · ΔG3 mean (.4/.5/.6): -2.45 pp



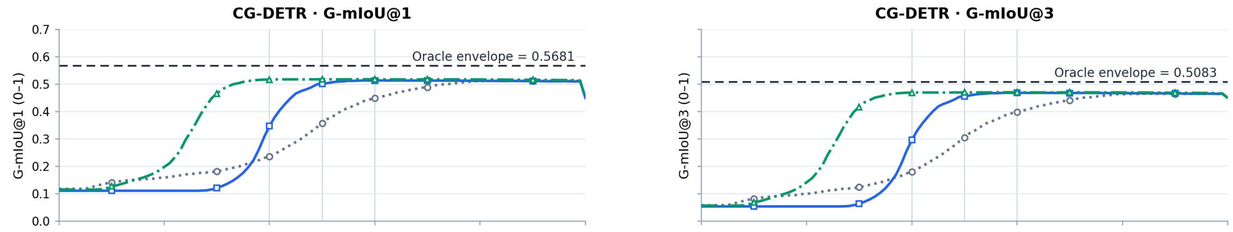
Baseline GMR | mAP 12.65% · DQ-CGPv3 GMR | mAP 13.93% · DQ-CGPv3 + GEQ Full | mAP 14.20%

ΔmAP (DQ-CGP vs Base): +1.28 pp · ΔG1 mean (.4/.5/.6): +0.23 pp · ΔG3 mean (.4/.5/.6): -0.43 pp



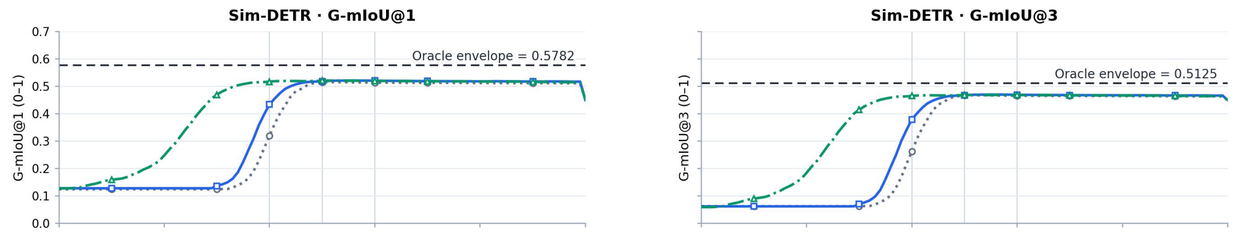
Baseline GMR | mAP 14.67% · DQ-CGPv3 GMR | mAP 12.49% · DQ-CGPv3 + GEQ Full | mAP 13.17%

ΔmAP (DQ-CGP vs Base): -2.18 pp · ΔG1 mean (.4/.5/.6): +10.68 pp · ΔG3 mean (.4/.5/.6): +11.20 pp



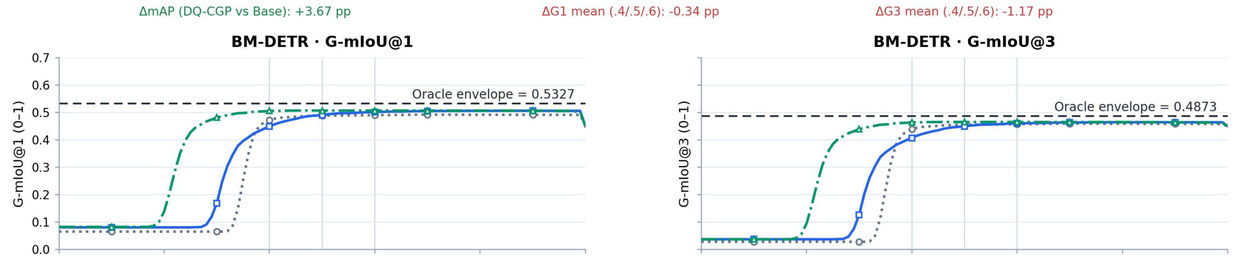
Baseline GMR | mAP 19.18% · DQ-CGPv3 GMR | mAP 19.62% · DQ-CGPv3 + GEQ Full | mAP 19.50%

ΔmAP (DQ-CGP vs Base): +0.44 pp · ΔG1 mean (.4/.5/.6): +4.20 pp · ΔG3 mean (.4/.5/.6): +4.05 pp



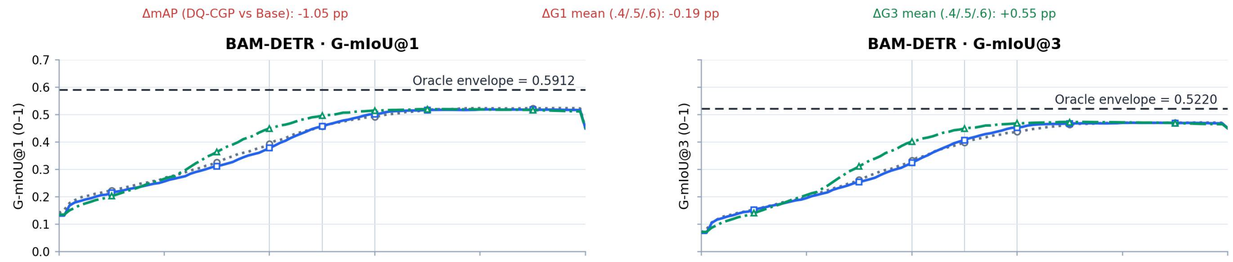
Baseline GMR | mAP 3.65% · DQ-CGPv3 GMR | mAP 7.32% · DQ-CGPv3 + GEQ Full | mAP 7.32%

ΔmAP (DQ-CGP vs Base): +3.67 pp · ΔG1 mean (.4/.5/.6): -0.34 pp · ΔG3 mean (.4/.5/.6): -1.17 pp



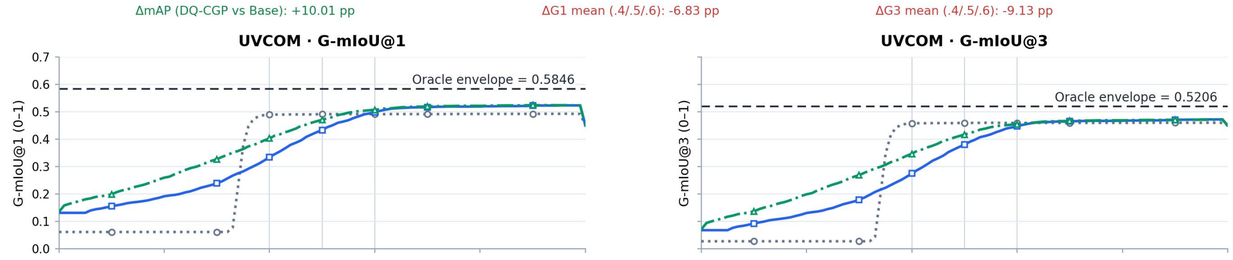
Baseline GMR | mAP 20.62% · DQ-CGPv3 GMR | mAP 19.57% · DQ-CGPv3 + GEQ Full | mAP 20.74%

ΔmAP (DQ-CGP vs Base): -1.05 pp · ΔG1 mean (.4/.5/.6): -0.19 pp · ΔG3 mean (.4/.5/.6): +0.55 pp



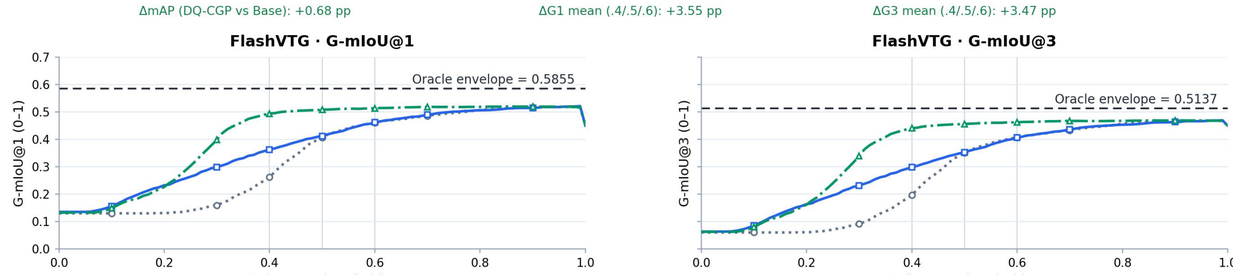
Baseline GMR | mAP 6.40% · DQ-CGPv3 GMR | mAP 16.41% · DQ-CGPv3 + GEQ Full | mAP 17.12%

ΔmAP (DQ-CGP vs Base): +10.01 pp · ΔG1 mean (.4/.5/.6): -6.83 pp · ΔG3 mean (.4/.5/.6): -9.13 pp



Baseline GMR | mAP 20.43% · DQ-CGPv3 GMR | mAP 21.11% · DQ-CGPv3 + GEQ Full | mAP 20.87%

ΔmAP (DQ-CGP vs Base): +0.68 pp · ΔG1 mean (.4/.5/.6): +3.55 pp · ΔG3 mean (.4/.5/.6): +3.47 pp



Oracle envelope: max of plotted methods with GT presence and fixed candidates. Guides at tau in {0.4, 0.5, 0.6}.